Exp No: 13	Implement your own ping program

AIM:

To implement our own ping program in Python that checks whether a host is reachable and displays the reply time of each attempt.

ALGORITHM / PROCEDURE:

1. Read the host name or the IP address from the user and convert it into an IP address using `socket.gethostbyname()`. If the host is invalid, display *Invalid host/IP address*.
2. Repeat four times: note the start time and try to open a TCP connection to the host on port 80 with `socket.create_connection((ip, 80), timeout=2)`.
3. If the connection is made, note the end time, display *Reply from <ip>* with the round trip time in milliseconds (end time - start time) and close the socket.
4. If the connection fails within 2 seconds, display *Request timed out*.
5. Wait for one second between the attempts.

DIAGRAM:

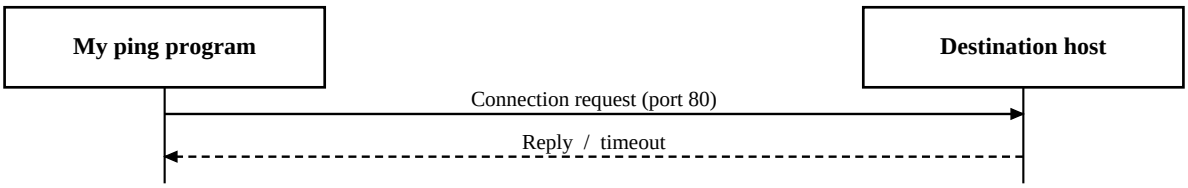

Fig 1: Working of the ping program

PROGRAM:

ping_program.py

```
import socket
import time
host = input("Enter host/IP address: ")
try:
    ip = socket.gethostbyname(host)
    print("Pinging", host, "[", ip, "]")
    for i in range(4):
        start = time.time()
        try:
            s = socket.create_connection((ip, 80), timeout=2)
            end = time.time()
            print("Reply from", ip, ": time =", round((end - start) * 1000, 2), "ms")
            s.close()
        except:
            print("Request timed out.")
        time.sleep(1)
except:
    print("Invalid host/IP address.")
```

OUTPUT:

Terminal (Linux) - python3 ping_program.py (target: a local web server on port 80)

```
Enter host/IP address: 127.0.0.1
Pinging 127.0.0.1 [ 127.0.0.1 ]
Reply from 127.0.0.1 : time = 0.6 ms
Reply from 127.0.0.1 : time = 0.2 ms
Reply from 127.0.0.1 : time = 0.18 ms
Reply from 127.0.0.1 : time = 0.18 ms
```

Terminal (Linux) - python3 ping_program.py (invalid host)

```
Enter host/IP address: abc.invalid
Invalid host/IP address.
```

RESULT:

Thus our own ping program was implemented in Python. The program checked the reachability of a host and displayed the reply time of each attempt.